

FULL NAME: _____

BLOCK: _____

L55 P1

Lesson 5-5 · Period 1 · Do Now

Name: _____ Period: _____ Date: _____

Do Now

In business, the term profit is used to describe the difference between the money the business earns (revenue) and the money the business spends (cost).

[IMAGE: Illustration of a man grooming a bulldog, with a sign reading "GROOMING 25^{perpet}".]

A. Let x represent the number of pets groomed in a month at Grooming USA. Define a revenue function for the business.

B. Materials and labor for each pet groomed cost 15. *The business also has fixed costs of 1,000 each month.* Define a cost function for this business.

C. Last month, Grooming USA groomed 95 pets. Did they earn a profit? What would the profit be if the business groomed 110 pets in a month?

D. Generalize Explain your procedure for calculating the profit for Grooming USA. Suppose you wanted to calculate the profit for several different scenarios. How could you simplify your process?